

SHREYAS ACHARYA

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EDUCATION

University of Maryland, College Park <i>MEng in Robotics</i>	Aug 2023 – May 2025 <i>College Park, MD</i>
Savitribai Phule Pune University <i>B.E in Computer Engineering</i>	Aug 2019 – May 2023 <i>Pune, IN</i>

SKILLS

Languages: Python, C++, MATLAB, SQL
Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, ROS2
Platforms & Tools: Linux, Hugging Face, RESTful APIs, Docker, Kubernetes, GitHub Actions, CI/CD, SolidWorks
Software Concepts: UML, Software Architecture, Agile Development, GPU Programming
Relevant Coursework: Multimodal Foundation Models, Computer Vision, Software Design, Data Structures

EXPERIENCE

techR Business Solutions <i>Machine Learning Engineer</i>	Jan 2023 – July 2023 <i>Pune, IN</i>
- Developed and deployed a real-time video processing pipeline to capture and classify over 400 distinct human actions, achieving over 77% accuracy. - Integrated a user-friendly Flask-based web interface for remote monitoring, featuring an interactive display and automated alerts that flagged suspicious activities with 96% precision.	

PROJECTS

Temporal Coherence Evaluation for Multimodal Foundation Models <i>Python, PyTorch, Hugging Face, CLIP</i>	Nov 2024
- Developed a robust evaluation framework to assess temporal reasoning in video-language models using semantic similarity (BERTScore), numeric distance, and a novel reference-free metric (CLIPGain). - Demonstrated significant gains in temporal understanding with increased frame context; semantic alignment improved by 6.2% (BERTScore) and numerical prediction accuracy by 55% across 200+ video samples. - Benchmarked six multimodal foundation models, revealing up to 14× improvement in cross-frame consistency (CLIPGain) for models leveraging full video context.	
Multiview Structure from Motion for 3D Scene Reconstruction <i>Python, OpenCV, NumPy, SciPy</i>	May 2024
- Designed a 3D reconstruction pipeline to accurately reconstruct scenes from multiple images, employing SIFT/FLANN-based matching for enhanced accuracy across synthetic and real-world datasets. - Incorporated custom Bundle Adjustment optimization to jointly refine 3D points, reducing reprojection errors by 10x across all datasets while preserving structural integrity, yielding high-fidelity 3D structures.	
Homography-Net: Deep Learning-Based Image Alignment & Panorama Stitching <i>Python, TensorFlow, NumPy, Keras</i>	March 2024
- Designed a deep neural network (VGG-19 backbone) to estimate homography transformations between image pairs, achieving an L2 pixel error of 5 through two-stage supervised training. - Implemented both supervised and unsupervised pipelines, with the supervised model reducing prediction error by 68%, enabling precise planar alignment across diverse scenes. - Built a traditional vision-based panorama stitching system using Shi-Tomasi corner detection and RANSAC, aligning up to 5 images with 98.4% feature match accuracy.	
Pb-lite Edge Detection: Multi-Cue Gradient-Based Boundary Detection <i>Python, TensorFlow, Scikit-learn, SciPy</i>	Jan 2024
- Developed an enhanced edge detection algorithm to accurately detect edges by combining brightness, color, and texture gradients across multiple scales, inspired by the classical Pb (probability of boundary) approach. - Improved edge continuity by approximately 30% through iterative refinement, significantly enhancing the detection of fine structures and preserving object boundaries with high spatial fidelity.	